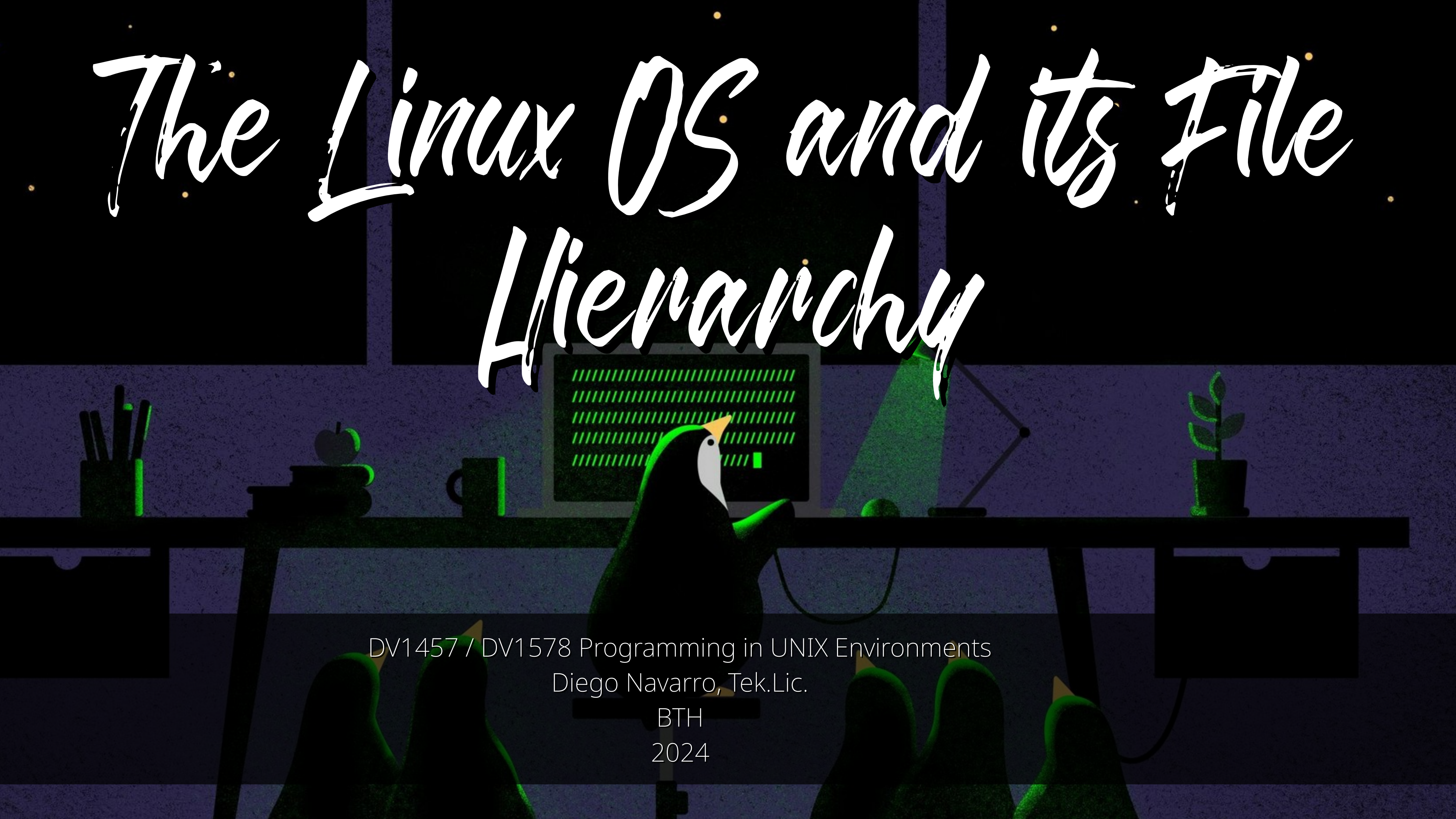
Linus Torvalds, Creator of Linux and GIT – The Linux Foundation



The Linux OS and its File Hierarchy

DV1457 / DV1578 Programming in UNIX Environments

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Program

- The GNU+Linux OS.
- Components of the Linux OS.
- The Linux file hierarchy.

The GNU+Linux OS

GNU+Linux is an operative system.

But..
What is an operative system?

Operative Systems

- “Imagine your computer hardware as individual instruments in an orchestra and applications as the musicians who play them. The operating system is the conductor, ensuring each instrument (hardware) works correctly and that the musicians (applications) have access to the right tools and resources to create beautiful music”.

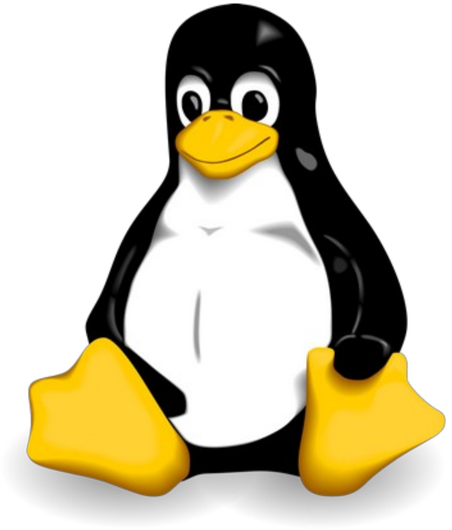
Gemma 2

Operative Systems

- An operating system is a collection of software programs that manage hardware resources and provide a standardized platform for developing an executing applications.

The GNU+Linux OS

- The GNU+Linux (Linux for short) is the term that refers to the family of operative systems that are based on the Linux kernel and the GNU project software packages.



The GNU+Linux OS

- The Linux OS implements a philosophy called “**everything is a file**”.
  - Every object in the Linux environment is treated as a file (i.e. hard drives, keyboards, terminal commands, network connections, etc).
  - Provide a **unified interface** to manipulate objects (e.g. open/close, read/write).
  - Allows the creation of customized data manipulation or interaction through **redirection and pipes** (i.e. using the output of a file as the input of another).

What are the advantages of this approach?

The GNU+Linux OS

- The "**everything is a file**" philosophy bring 3 major advantages:
  1) **Simplicity**: having a unified interface for ever object in the system makes easier to track, implement, and understand the OS.
  2) **Extensibility**: New features can be implemented without requiring to modify existing ones.
  3) **Portability**: Linux scripts work (most of the times seamlessly) among its many distributions.

The GNU+Linux OS

- There are a plethora of Linux variations (a.k.a. distributions), **customized** for specific **tasks**, **software**, or to follow a specific **philosophy** (e.g. to use or not proprietary software).
  - I.M.O. Linux distributions can be classified in 4 major branches.

The GNU+Linux OS

- The Slackware Branch:
  - Based on Slackware OS, the oldest on-going Linux distribution (since 1993).
  - Very minimal systems, focused on stability and simplicity.
  - Only installs software from source-code.
  - Not very popular and it targets experienced users.

The GNU+Linux OS

- The Slackware Branch:
  - Examples: Slackware, Slax.

The GNU+Linux OS

- **The Debian Branch**:
  - Based on Debian OS.
  - **The largest** (and perhaps most influential) branch of distributions.
  - Focus on **stability, reliability, free and open source software adherence, and community-driven development**.
  - **Test** software packages **extensively** before releasing them.
  - Targets a **general audience**.

The GNU+Linux OS

- The Debian Branch:
  - Examples: Debian, Ubuntu, Linux Mint, Raspberry Pi OS (Raspbian).

The GNU+Linux OS

- The Red Hat Enterprise Linux (REHL) Branch:
  - Based on Fedora.
  - The powerhouse of enterprise and server-space Linux.
  - Focus on stability, security, compatibility, and long-term commercial support.
  - Target enterprises and optimized for professional work and safety-critical systems.

The GNU+Linux OS

- The Red Hat Enterprise Linux (REHL) Branch:
  - Examples: RHEL, Fedora, CentOS, Rocky Linux, Alma Linux.

The GNU+Linux OS

- The Arch Branch:
  - Based on Arch Linux.
  - Focuses on bare-bone systems and in-depth customization.
  - Community driven project.
  - Targets experienced users.

The GNU+Linux OS

- The Arch Branch:
  - Examples: Arch Linux, Manjaro.

The GNU+Linux OS

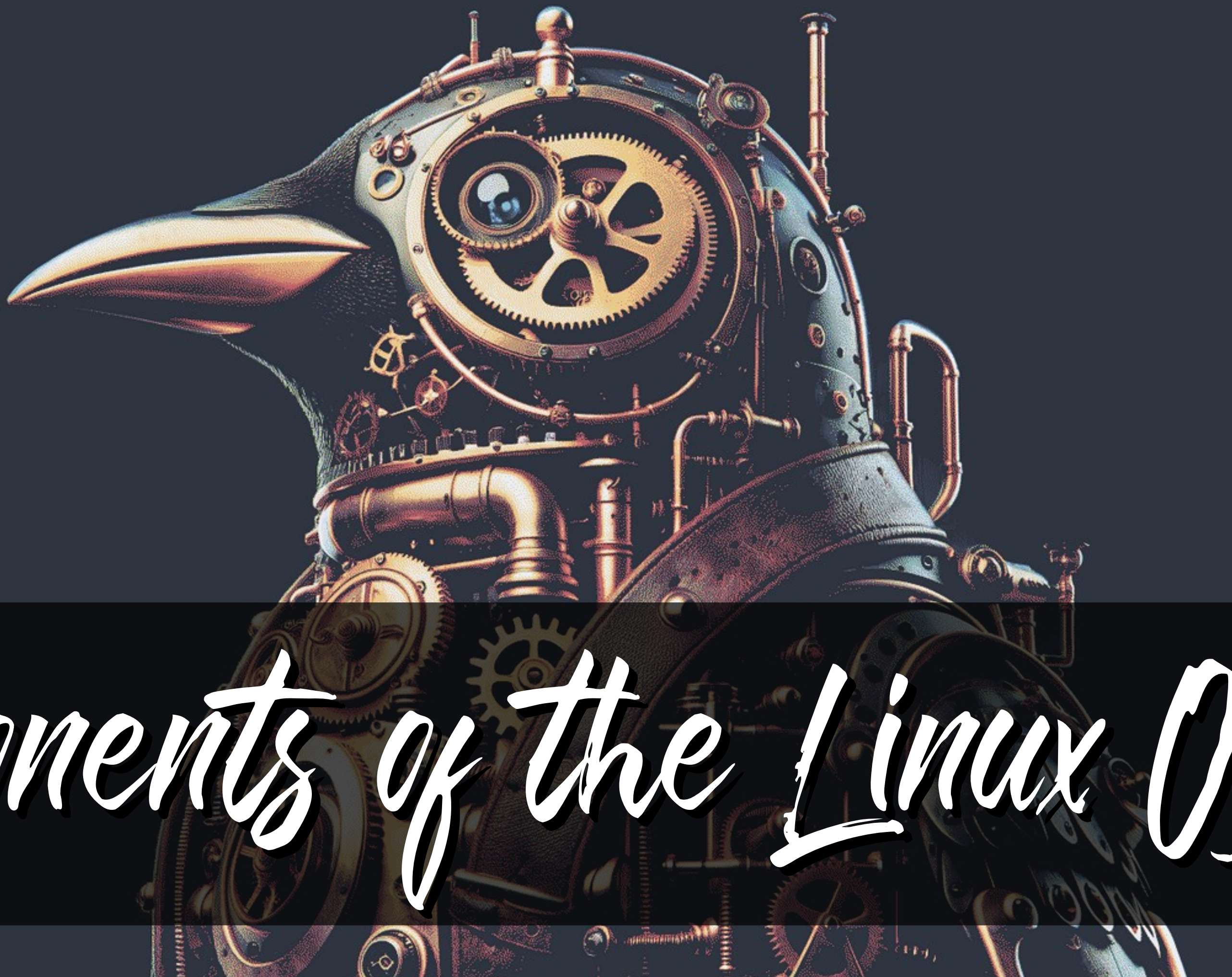
- **Other noteworthy distributions**:
  - OpenSUSE: The other powerhouse in the **enterprise** and **server space**.
  - Gentoo: Granular **control** of the OS for **performance optimization**.
  - Kali Linux: Strong focus on **cybersecurity** and **ethical hacking**.
  - NixOS: Immutable system focused on reproducibility and stability.

The GNU+Linux OS

- Linux distributions are a major rabbit hole.
  - There are all kinds of distributions out there, explore at your own risk!

- Despite this, distributions are more a convenience than anything else.
  - You can build any distribution, staring from any distribution.

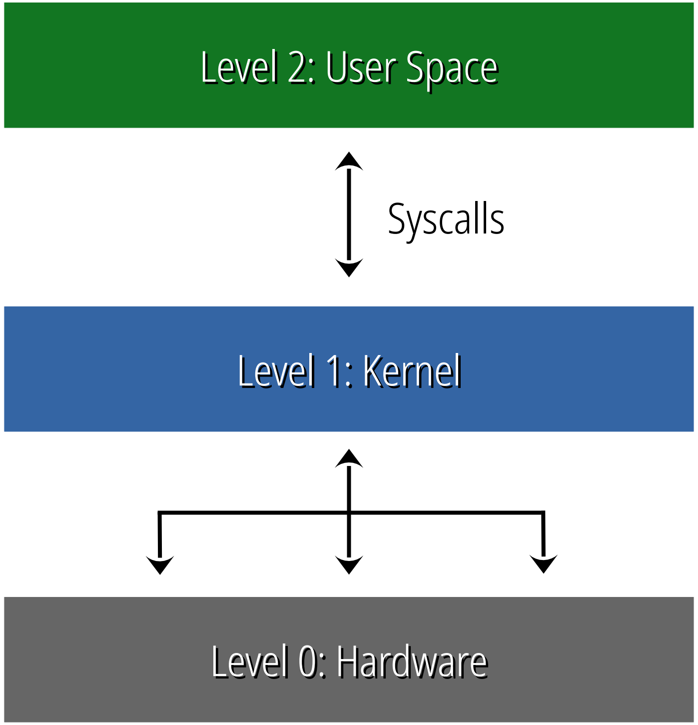
What components are needed to build an operative system?



Components of the Linux OS

Components of the Linux OS

- The Linux architecture is commonly represented in a 3-level interconnected structure.



Components of the Linux OS

- Level 2: User Space.
  - Set of components that enable user I/O and interaction.
  - Elements in the user space emit a signal (i.e. system call or "syscall") for the Kernel to manage, allocate resources, and process.
  - It is composed (primarily) by: Applications, Package Manager, Desktop environment, Display Server, Terminal, Shell, and System utilities.

Components of the Linux OS

- **Applications:** The user programs running in the system.
- **Package Manager:** Tool that automates i**nstalling, updating, removing, and managing the software packages** in the system. It is distribution - dependent:
  - Slackware branch: pkgtool.
  - Debian branch: APT.
  - REHL branch: DNF, YUM.
  - ARCH branch: Pacman.

Components of the Linux OS

- Desktop environment: collection of interconnected software components that create the system GUI and the system interaction methods.
  - GNOME: focuses on modern looks and simplicity (of design and interaction).
  - KDE: focuses on scalability, productivity, and customization.
  - XFCE: focuses on resource optimization (used commonly in older or resource-limited hardware).

Components of the Linux OS

- Display server: Software component in charge of the managing and rendering of graphic output of the system.
  - This includes window managing (manipulation of window geometry), rendering (processing and frame-buffering graphical data), and the display output (i.e. create signals from buffered data according to the required protocol – VGA, HDM, DP, etc.-).
- Two main display servers:
  - X11: mature, stable, well-supported.
  - Wayland: newer, modular, focused on improving performance.

Components of the Linux OS

- **Terminal (emulator): text-based interface** that allows the user to **input commands to control the OS** directly.
- **Shell:** program that runs inside the terminal, that acts as the **command interpreter** for the execution of specified instructions.
  - Bash (Bourne again shell).
  - CSH (C shell).
  - KSH (Korn shell).

Components of the Linux OS

- **System utilities:** Set of standardized tools to manipulate and interact directly with the OS.
  - File management: ls, cd, mkdir, touch.
  - Text manipulation: cat, grep, sort.
  - Process management: ps, top, kill.
  - Network: ping, curl, wget.
  - System information: uname, whoami.

Components of the Linux OS

- Level 1: Kernel.
  - Monolithic component in charge of receiving syscalls, and to allocate and manage hardware resources for their execution.
  - A single syscall can be divided into multiple hardware request, depending on the needs of the instruction.

Components of the Linux OS

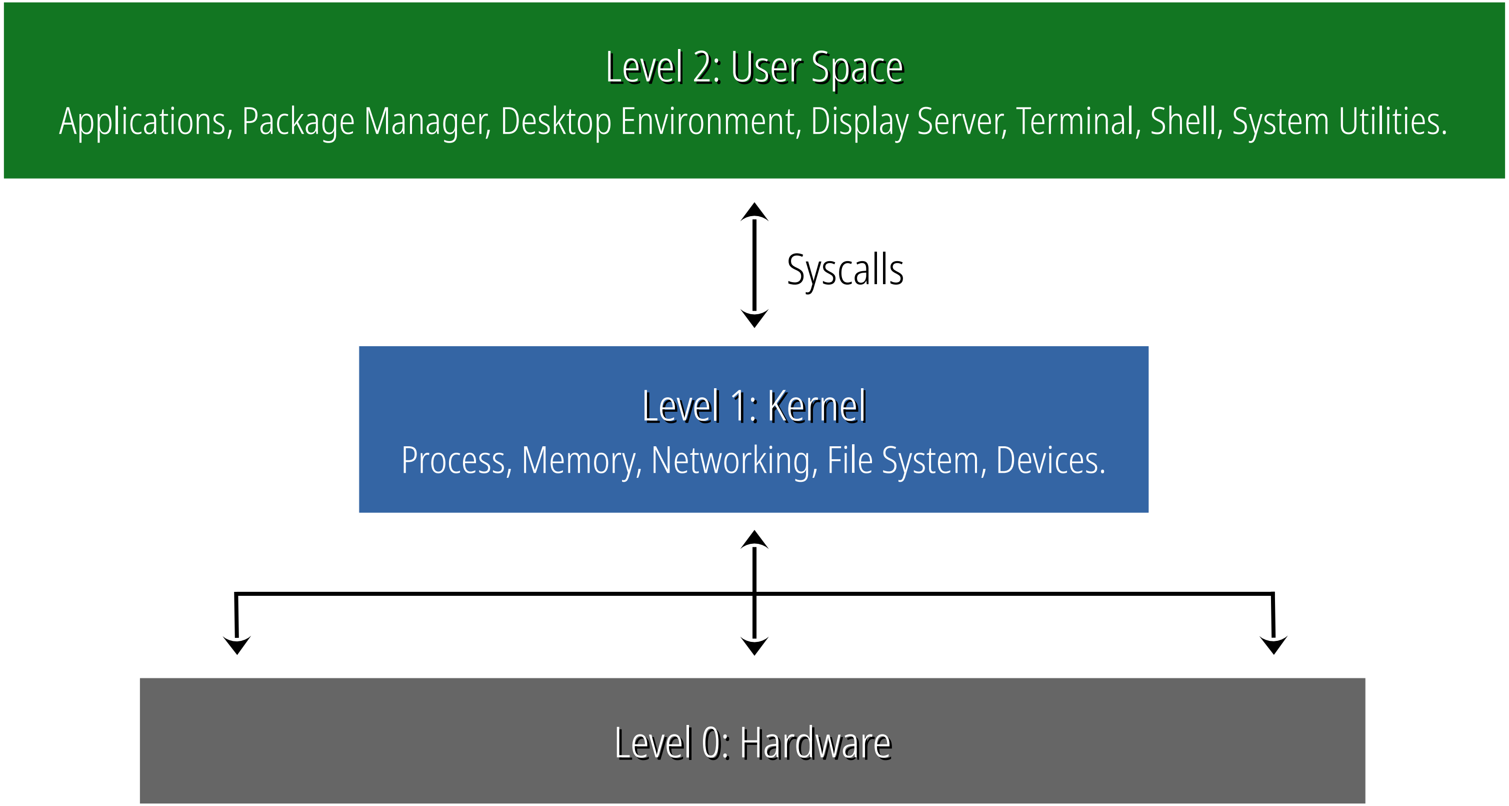
- Level 1: Kernel.
  - The Linux kernel features 5 main interfaces:

1) Process: threads and processing management (CPU/GPU).
2) Memory: allocates and maps system memory (RAM, ROM, Cache).
3) Networking: registers and manages the network stack (NIC).
4) File system: manages system files and enables file creation and deletion (HDD, SSD, NVME).
5) Devices: manages external devices and its drivers (e.g. Keyboard)

Components of the Linux OS

- **Level 0: Hardware**.
  - All the physical components in the computer running the system.
  - CPU/GPU, RAM/ROM, NIC, HDD/SSD/NVME, External devices (e.g. keyboard).

Components of the Linux OS

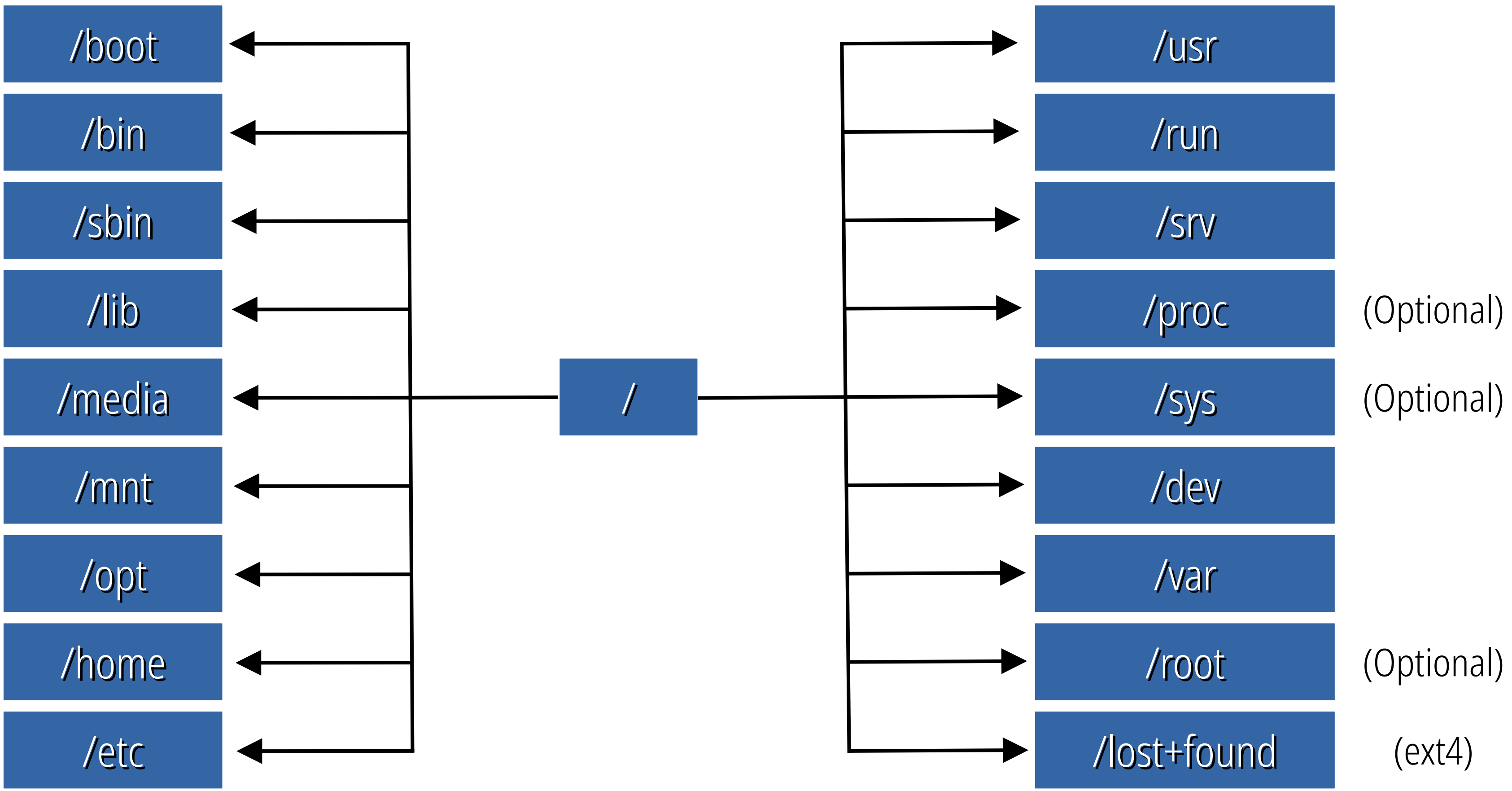



The Linux File Hierarchy

The Linux File Hierarchy

- Standardized structure for organizing files an directories in the Linux system.
  - It follows the Filesystem Hierarchy Standard (FHS).
- Consistent among UNIX / Linux distributions.
  - Every distribution may modify/expand over the FHS.

The Linux File Hierarchy



The Linux File Hierarchy

- Root Directory (/): contains all files and directories of the Linux system.
- /bin: Stores essential system utilities (e.g. ls, cat, vi, ps, etc.)
- /sbin: Stores system administration utilities (e.g. init, ipconfig, etc.)
- /lib and /usr/lib: stores libraries required by the kernel and the user space program respectively (i.e. .so files).

The Linux File Hierarchy

- /media and /mnt: Stores the mount points for removable devices (e.g. USB drives) or physical storage (e.g. HDD, SSD, etc.).
- /opt: Stores optional application packages, that are not part of the OS (e.g. plugins).
- /home: Stores home directories for user accounts.
- /etc: stores system-wide configuration files (e.g. fstab, ssh_config, etc.)

The Linux File Hierarchy

- /usr: Stores application dependencies that are not required during the system boot.
- /run: Stores files and data created during runtime and that is needed in the current boot session (e.g. open applications/files, sockets, etc.).
- /srv: Stores data from network services (HTTP, FTP, SSH, DNS, etc.).
- /dev: Stores data for all devices (i.e. hardware components) connected to system.

The Linux File Hierarchy

- /var: Stores variable data files (e.g. log files, databases).
- /proc (optional): Stores memory-mapped files that access the current kernel data structures (e.g. system processes, memory usage, etc.).
- /sys (optional): Stores read-only data of the kernel interfaces (CPU usage, network stack, etc.).
- /root (optional): Stores the home directory of the root user.
- /lost+found (ext4): Stores corrupted files or files with errors when mounting the drive.


Closing Remarks

Closing Remarks

- We have just scratch the surface!
  - We will focus on User Space and the Shell onward.
- Mastering Linux require time and practice.
- There is no other computational tool like GNU+Linux!
  - Not just a tool but a community as well! ❤️
- Additional material added in Canvas.


WHAT?